DATABASES

# Cassandra Visual Guide

Wide-column storage, tunable consistency, gossip -- built for write scale.

**ANIMATED WALKTHROUGH**

Step through pages to play the animation

STEP 1/6

# The Wide-Column Model

Partition key decides the node; clustering key orders rows inside.

```
CREATE TABLE readings (
  sensor_id text,                 -- PARTITION KEY: which node
  ts timestamp,                   -- CLUSTERING KEY: order within partition
  temp float, humidity float,
  PRIMARY KEY (sensor_id, ts) );
```

**Partition: sensor-42**

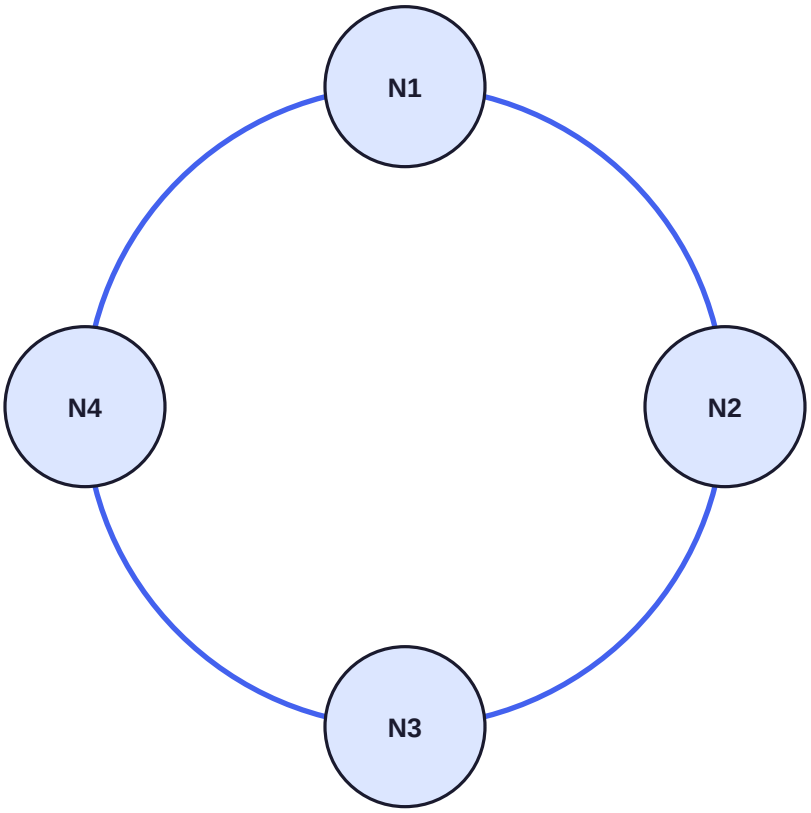
ts rows stored sorted together -- one partition = one fast sequential read

Design rule: model around queries. Each query should ideally hit ONE partition.

STEP 2/6

# Partitioning & the Token Ring

Every node owns token ranges; data is hashed onto the ring.




hash(partition key) -> token -> owning node. Vnodes split each node into many small ranges.

Replication factor 3: each partition lives on 3 nodes around the ring -- no single point of loss.

STEP 3/6

# Tunable Consistency

Choose per query: how many replicas must agree?

**ONE**

fastest, weakest -- 1 replica ack

**QUORUM**

majority -- the sane default

**LOCAL_QUORUM**

majority within the datacenter

**ALL**

every replica -- slowest, strongest

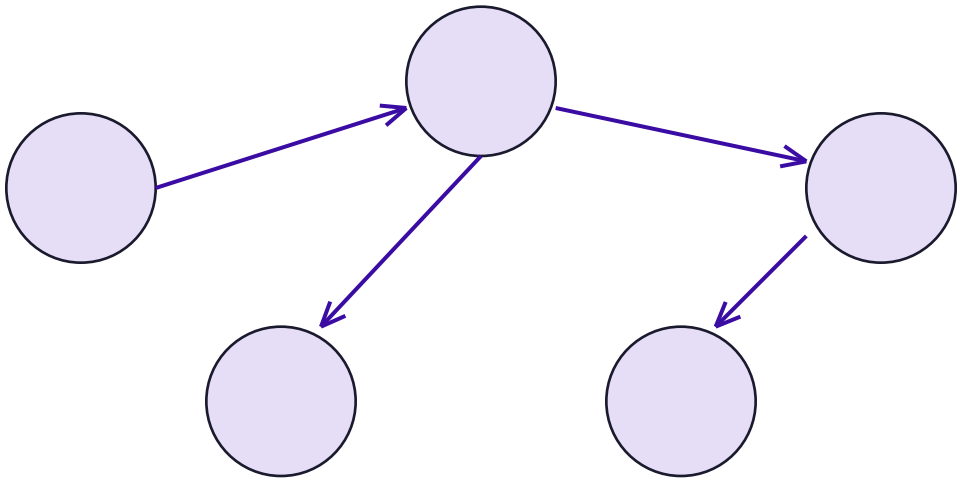
**The quorum rule: R + W > N**

read replicas + write replicas > replication factor = strong consistency

STEP 4/6

# Gossip Protocol

No master: nodes gossip state to each other.

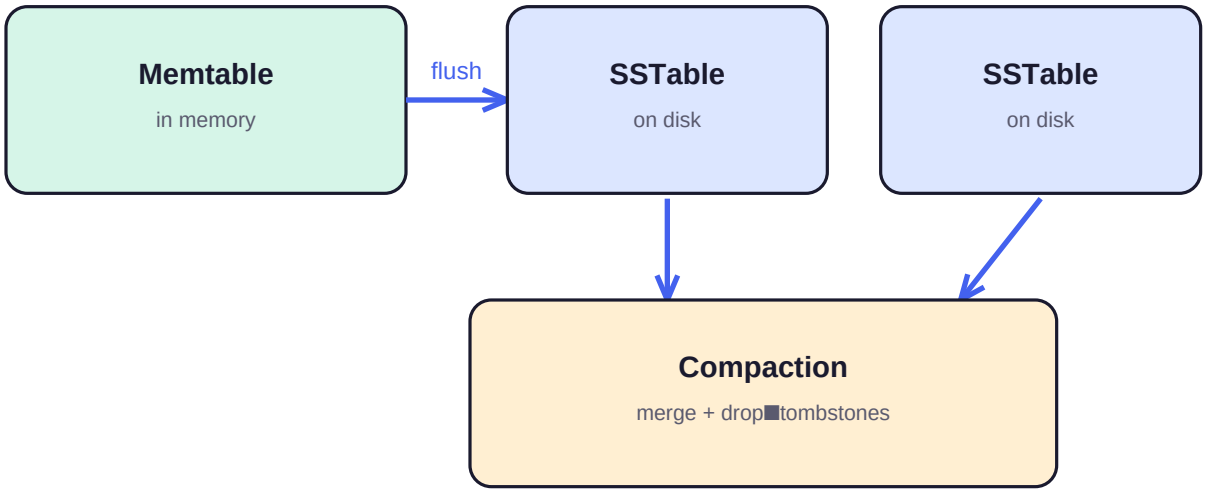


- Every second, each node gossips its state (up/down, load, schema) to random peers.
- Failure detection is decentralized: silence for a timeout = suspected down.
- Seed nodes bootstrap new joiners; after that every node is equal.

STEP 5/6

# Compaction

Writes are fast because they are append-only -- compaction cleans up.




- LSM-tree: writes hit memtable, flush to immutable SSTables -- never random writes.
- Compaction merges SSTables, evicts deleted data (tombstones) and expired TTLs.
- Strategies: STCS (default), LCS (read-heavy), TWCS (time series). Pick per table.

STEP ✓/✓

# Key Takeaways

## Cassandra -- Remember

- Model per query: partition key picks the node, clustering key orders rows inside it.
- Consistent hashing on a token ring spreads data; RF=3 keeps every partition on three nodes.
- Consistency is tunable per query -- QUORUM is the sane default; R + W > N gives strong reads.
- Gossip replaces the master: decentralized membership and failure detection.
- LSM writes are append-only fast; compaction merges SSTables and reclaims tombstones.

**Keep exploring**

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev